LUCIDMSK CME PROGRAM

MODULE 10 — POST-TEST

MSK Soft Tissue & Bone Tumors: A Systematic Imaging Approach

Soft Tissue Tumors · 10 Questions · 1.5 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 **A 25-year-old with rapidly enlarging thigh mass. MRI shows intramuscular lesion with peripheral ossification and no enhancement. Most likely diagnosis:**

A	Liposarcoma
B	Myositis ossificans (USP6 family)
C	Osteosarcoma
D	Lymphoma

2 **BNCT diagnostic criteria (ALL required). Which combination is correct?**

A	T1 high, T2 low, enhancement present, central vertebral
B	T1 low, T2 high, NO enhancement, central vertebral, ± mild sclerosis
C	Lytic lesion, T2 high, soft tissue mass, enhancement
D	T1 high, T2 low, no enhancement, eccentric cortex-based

3 **Key MRI discriminating feature: lipoma vs. well-differentiated liposarcoma (WDLPS):**

A	Size (>10 cm = sarcoma)
B	Deep vs. superficial location
C	Non-adipose (thick) internal septa or non-fatty nodular components
D	T1 signal of the fat component

4 **Periarticular lobulated mass with low T1 AND low T2 signal in the knee. Most likely:**

A	Ganglion cyst
B	Lipoma arborescens
C	PVNS / TSGCT
D	Baker cyst

5 Aggressive periosteal lesion in 15-year-old: Codman triangle + "sunburst" periosteal reaction + metaphyseal location. Most likely:

- **A** Ewing sarcoma
- **B** Conventional osteosarcoma
- **C** Osteoid osteoma
- **D** Giant cell tumor

6 Goutallier Grade 3 fatty infiltration in the gluteus medius. Most consistent with:

- **A** Normal aging
- **B** Greater trochanteric pain syndrome with chronic abductor tear
- **C** Acute gluteus medius strain
- **D** Piriformis syndrome

7 Typical BNCT vertebral lesion — correct management recommendation:

- **A** CT-guided biopsy immediately
- **B** PET-CT for staging
- **C** No biopsy — imaging surveillance appropriate; biopsy NOT recommended
- **D** Surgical resection

8 The "zonal phenomenon" of myositis ossificans on CT:

- **A** Dense central mineralization; lucent periphery
- **B** Peripheral mature mineralization; less organized soft center (outside-in pattern)
- **C** Uniform amorphous calcification throughout
- **D** Cortical disruption with central necrosis

9 **Periosteal-based lesion arising from the posterior distal femur in a 35-year-old, WITHOUT intramedullary involvement. Most likely:**

A	Conventional osteosarcoma
B	Parosteal osteosarcoma (low-grade)
C	Periosteal chondrosarcoma
D	Giant cell tumor

10 **Most sensitive sequence for hemosiderin detection in PVNS/TSGCT:**

A	T1 fat-sat
B	T2 fat-sat (STIR)
C	Gradient echo (GRE) — "blooming" artifact
D	Post-contrast T1 fat-sat

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	B	B	C	C	B	B	C	B	B	C

Q1 **A 25-year-old with rapidly enlarging thigh mass. MRI shows intramuscular lesion with peripheral ossification and no enhancement. Most likely diagnosis:** **Answer: B**

A Liposarcoma
✓ **Myositis ossificans (USP6 family)**
C Osteosarcoma
D Lymphoma

EXPLANATION Myositis ossificans = USP6 family: young patient + intramuscular + peripheral-to-central ossification ("zonal phenomenon") + no enhancement of the ossified rim. Opposite pattern from osteosarcoma.

Q2 **BNCT diagnostic criteria (ALL required). Which combination is correct?** **Answer: B**

A T1 high, T2 low, enhancement present, central vertebral
✓ **T1 low, T2 high, NO enhancement, central vertebral, ± mild sclerosis**
C Lytic lesion, T2 high, soft tissue mass, enhancement
D T1 high, T2 low, no enhancement, eccentric cortex-based

EXPLANATION BNCT criteria: T1 hypointense, T2 hyperintense, NO enhancement, central vertebral, ± mild sclerosis, ± multifocal. Any frank lysis, extraosseous mass, or avid enhancement → raises chordoma. Biopsy NOT recommended for typical BNCT.

Q3 **Key MRI discriminating feature: lipoma vs. well-differentiated liposarcoma (WDLPS):** **Answer: C**

A Size (>10 cm = sarcoma)
B Deep vs. superficial location
✓ **Non-adipose (thick) internal septa or non-fatty nodular components**
D T1 signal of the fat component

EXPLANATION Key discriminator: non-fatty nodular components or thick (>2 mm) irregular septa = WDLPS until proven otherwise. Pure lipoma = thin (<2 mm) regular septa, homogeneous fat. Size and depth alone are insufficient.

Q4 **Periarticular lobulated mass with low T1 AND low T2 signal in the knee. Most likely:** **Answer: C**

A Ganglion cyst
B Lipoma arborescens
✓ **PVNS / TSGCT**
D Baker cyst

EXPLANATION PVNS / TSGCT = low T1 AND low T2 signal due to hemosiderin (paramagnetic). GRE sequences show blooming artifact. Ganglion = high T2. Lipoma = T1 bright. Baker cyst = T2 bright fluid.

Q5 **Aggressive periosteal lesion in 15-year-old: Codman triangle + "sunburst" periosteal reaction + metaphyseal location. Most likely:** **Answer: B**

A Ewing sarcoma

✓ **Conventional osteosarcoma**

C Osteoid osteoma

D Giant cell tumor

EXPLANATION Osteosarcoma: Codman triangle + sunburst/spiculated periosteal reaction + metaphyseal location in adolescent. Ewing = diaphyseal + onion-skin periosteal reaction. Both may show Codman triangle.

Q6 **Goutallier Grade 3 fatty infiltration in the gluteus medius. Most consistent with:** **Answer: B**

A Normal aging

✓ **Greater trochanteric pain syndrome with chronic abductor tear**

C Acute gluteus medius strain

D Piriformis syndrome

EXPLANATION Goutallier Grade 3 (fat = muscle) in the gluteus medius indicates chronic atrophy from long-standing abductor tendon tear — the "rotator cuff of the hip." Hallmark of chronic GTPS with established tendon failure.

Q7 **Typical BNCT vertebral lesion — correct management recommendation:** **Answer: C**

A CT-guided biopsy immediately

B PET-CT for staging

✓ **No biopsy — imaging surveillance appropriate; biopsy NOT recommended**

D Surgical resection

EXPLANATION Typical BNCT does NOT require biopsy. Biopsy may miss the diagnosis or cause morbidity. Surveillance is recommended. Atypical features (lysis, soft tissue mass, enhancement) → atypical notochordal tumor → biopsy warranted.

Q8 **The "zonal phenomenon" of myositis ossificans on CT:** **Answer: B**

A Dense central mineralization; lucent periphery

✓ **Peripheral mature mineralization; less organized soft center (outside-in pattern)**

C Uniform amorphous calcification throughout

D Cortical disruption with central necrosis

EXPLANATION Myositis ossificans zonal phenomenon: mature peripheral ossification (organized cortical bone) with immature soft center — OUTSIDE-IN pattern. Osteosarcoma is the opposite (central matrix, disorganized periphery). Diagnostic on CT.

Q9 **Periosteal-based lesion arising from the posterior distal femur in a 35-year-old, WITHOUT intramedullary involvement. Most likely:** **Answer: B**

A Conventional osteosarcoma

✓ **Parosteal osteosarcoma (low-grade)**

C Periosteal chondrosarcoma

D Giant cell tumor

EXPLANATION Parosteal osteosarcoma: posterior distal femur (90%), periosteal-based (NOT intramedullary), low-grade, densely ossified, broad base on CT. Conventional osteosarcoma is intramedullary. GCT is epiphyseal/subarticular.

Q10 **Most sensitive sequence for hemosiderin detection in PVNS/TSGCT:** **Answer: C**

A T1 fat-sat

B T2 fat-sat (STIR)

✓ **Gradient echo (GRE) — "blooming" artifact**

D Post-contrast T1 fat-sat

EXPLANATION GRE sequences are most sensitive for hemosiderin — "blooming" artifact amplifies local field inhomogeneity. Hemosiderin is suppressed on spin-echo sequences. This is the hallmark of PVNS/TSGCT and hemophilic arthropathy.